Double Spend Probability

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1 Parameters

- λ - global block creation rate - assume $\lambda = 0.1$ blocks/min in Bitcoin
- α - fraction of attacker hashpower
- T - time since transaction broadcast - lower bounded by time since transaction included in a block
- N - number of main chain blocks added since transaction broadcast

2 Assumptions

- No attacker premine - attacker begins the double spend attempt when the transaction is broadcast
- Transaction included in the next block after it was broadcast

3 Some Equations

Probability the attacker creates k blocks over a window of time T is:

$$P_{\text{attacker_creates}}(k, T) = \frac{(\alpha\lambda T)^k e^{-\alpha\lambda T}}{k!}$$

The number of blocks created by a particular miner from the time when a particular transaction is broadcast is Poisson distributed. Note, calculating this precisely is tricky due to numerical issues, so I advise you to use a library for evaluating expressions of this form.

Since N represents the number of main chain blocks since a particular transaction broadcast, the probability the attacker is exactly i blocks ahead (if attacker is behind, i is negative) at the end of the time window is:

$$P_{\text{attacker_ahead_by_}i}(i, T, N) = \frac{(\alpha\lambda T)^{i+N} e^{-\alpha\lambda T}}{(i+N)!}$$

As an example, evaluating when $i = 2$ gives the probability the attacker finds $N + 2$ blocks, putting the attacker 2 blocks ahead of the main chain.

The probability a particular transaction is double spent assuming it is accepted *right now* is:

$$P_{\text{double_spend_if_accepted_right_now}}(T, N) = \sum_{i=-N}^{0} P_{\text{attacker_ahead_by_}i}(i, T, N) \left(\frac{\alpha}{1-\alpha}\right)^{-i+1} + 1 - \sum_{i=-N}^{0} P_{\text{attacker_ahead_by_}i}(i, T, N)$$

The left sum above corresponds to the probability of the attacker catching up sometime later (after transaction acceptance). The remainder of the probability corresponds to the attacker double spending immediately at transaction acceptance.